

MCQ Question Bank

Software

Python & Programming Concepts

1. In Python, what is the main difference between a list comprehension and a generator expression?

- A. List comprehensions are slower than generator expressions
- B. List comprehensions compute all values in memory immediately, while generator expressions yield items lazily on demand
- C. Generator expressions return a tuple, while list comprehensions return a list
- D. There is no difference in memory usage or execution speed

Correct Answer: B - List comprehensions compute all values in memory immediately, while generator expressions yield items lazily on demand

2. In Python object-oriented programming, which dunder method is responsible for instantiating a new object instance before initialization?

- A. `__init__`
- B. `__new__`
- C. `__create__`
- D. `__construct__`

Correct Answer: B - `__new__`

3. Which Python decorator is used to define a method that receives the class itself as its first argument rather than an instance?

- A. `@staticmethod`
- B. `@classmethod`
- C. `@property`
- D. `@abstractmethod`

Correct Answer: B - `@classmethod`

Software

SDLC & Agile Methods

1. In the Test-Driven Development (TDD) methodology, what is the correct sequential order of the development loop?

- A. Write Code -> Refactor -> Write Test
- B. Write Test (Red) -> Pass Test (Green) -> Refactor
- C. Refactor -> Write Test -> Write Code
- D. Write Test -> Document Code -> Deploy

Correct Answer: B - Write Test (Red) -> Pass Test (Green) -> Refactor

2. Which Agile Scrum event occurs at the end of a sprint to reflect on processes and discuss potential team improvements?

- A. Sprint Planning
- B. Daily Standup
- C. Sprint Review
- D. Sprint Retrospective

Correct Answer: D - Sprint Retrospective

3. What does the 'O' stand for in the SOLID principles of object-oriented design?

- A. Object-Oriented Principle
- B. Open/Closed Principle
- C. Operational Dependency Principle
- D. Overriding Method Principle

Correct Answer: B - Open/Closed Principle

Software

Cloud Computing & Architecture

1. Which Cloud service model provides virtualized computing infrastructure such as virtual machines, storage, and firewalls?

- A. SaaS (Software as a Service)
- B. PaaS (Platform as a Service)
- C. IaaS (Infrastructure as a Service)
- D. FaaS (Function as a Service)

Correct Answer: C - IaaS (Infrastructure as a Service)

2. What architectural pattern breaks down an application into loosely coupled, independently deployable services?

- A. Monolithic Architecture
- B. Microservices Architecture
- C. Layered (N-Tier) Architecture
- D. Event-Driven Monolith

Correct Answer: B - Microservices Architecture

3. What is a primary characteristic of Serverless / FaaS (Function as a Service) execution models?

- A. Servers must be manually provisioned and patched 24/7
- B. Code runs in response to events and scales automatically without managing underlying infrastructure
- C. Billing is based on fixed monthly reserved instances
- D. State is permanently stored inside the execution memory container

Correct Answer: B - Code runs in response to events and scales automatically without managing underlying infrastructure

4. In continuous deployment practices, what is the purpose of a Canary deployment?

- A. Rolling out software updates to 100% of servers at once
- B. Releasing new features incrementally to a small subset of users before full rollout
- C. Running tests exclusively in a local offline environment
- D. Automating database schema backup operations

Correct Answer: B - Releasing new features incrementally to a small subset of users before full rollout

Data Center

Virtualization & Hypervisors

1. What is the key structural difference between a Type-1 (Bare-Metal) and a Type-2 (Hosted) Hypervisor?

- A. Type-1 runs directly on hardware, while Type-2 runs on top of an existing host operating system
- B. Type-2 runs directly on hardware, while Type-1 requires a Linux kernel
- C. Type-1 can only run Windows virtual machines
- D. Type-2 hypervisors do not support virtual network interface controllers

Correct Answer: A - Type-1 runs directly on hardware, while Type-2 runs on top of an existing host operating system

2. What technology enables live migration of running virtual machines between physical hosts without downtime?

- A. Storage Snapshot
- B. vMotion / Live Migration
- C. Hot-Swapping
- D. Dynamic Host Configuration Protocol

Correct Answer: B - vMotion / Live Migration

3. In containerized data center workloads, what key operating system component is shared among all containers?

- A. Independent RAM allocation
- B. Host OS Kernel
- C. Dedicated BIOS/UEFI firmware
- D. Virtual CPU Socket

Correct Answer: B - Host OS Kernel

Data Center

Storage Architecture

1. Which Storage Area Network (SAN) protocol encapsulates SCSI commands into IP packets to operate over Ethernet networks?

- A. Fibre Channel (FC)
- B. iSCSI
- C. NFS
- D. SMB

Correct Answer: B - iSCSI

2. Which RAID configuration provides block-level striping with distributed parity across a minimum of 3 drives?

- A. RAID 0
- B. RAID 1
- C. RAID 5
- D. RAID 10

Correct Answer: C - RAID 5

3. In disaster recovery planning, what does Recovery Time Objective (RTO) measure?

- A. The maximum acceptable duration of system downtime following a failure
- B. The maximum acceptable amount of data loss measured in time
- C. The bandwidth required to replicate data across data centers
- D. The physical distance required between primary and backup sites

Correct Answer: A - The maximum acceptable duration of system downtime following a failure

Data Center

Facility Infrastructure & Power

1. How is Power Usage Effectiveness (PUE) calculated in data center efficiency metrics?

- A. Total Facility Energy divided by IT Equipment Energy
- B. IT Equipment Energy divided by Total Facility Energy
- C. Cooling Power divided by UPS Battery capacity
- D. Total Electrical Draw multiplied by Rack Density

Correct Answer: A - Total Facility Energy divided by IT Equipment Energy

2. In modern data center HVAC design, what is the purpose of Hot Aisle / Cold Aisle containment?

- A. To keep server cables separated from power lines
- B. To prevent hot exhaust air from mixing with cold intake air, improving cooling efficiency
- C. To isolate fire suppression gas within specific racks
- D. To create vacuum pressure inside server chassis

Correct Answer: B - To prevent hot exhaust air from mixing with cold intake air, improving cooling efficiency

3. What standard hardware interface provides out-of-band management capabilities for remotely monitoring hardware independent of the OS?

- A. IPMI (Intelligent Platform Management Interface)
- B. DHCP
- C. SNMP v1
- D. PXE Boot

Correct Answer: A - IPMI (Intelligent Platform Management Interface)

4. What does an N+1 redundancy model signify for critical data center components (e.g., UPS or Generators)?

- A. Double the required operating capacity is available
- B. There is one additional backup component available beyond the operational baseline requirements
- C. Components operate without any backup unit
- D. 100% of data center power is sourced from solar microgrids

Correct Answer: B - There is one additional backup component available beyond the operational baseline requirements

Cyber Security

Cryptography & Authentication

1. In asymmetric key cryptography, if User A wants to send an encrypted message to User B that only User B can read, which key should User A use?

- A. User A's Private Key
- B. User A's Public Key
- C. User B's Public Key
- D. User B's Private Key

Correct Answer: C - User B's Public Key

2. What is a core cryptographic property of secure hash functions like SHA-256?

- A. They are easily reversible using a master password
- B. They are one-way functions where a small change in input drastically changes the output digest
- C. They produce variable-length hash strings depending on input file size
- D. They rely on symmetric session key exchange protocols

Correct Answer: B - They are one-way functions where a small change in input drastically changes the output digest

3. Which factor category does a hardware TOTP security token (like YubiKey or Authenticator App) represent in Multi-Factor Authentication?

- A. Something You Know
- B. Something You Have
- C. Something You Are
- D. Somewhere You Stand

Correct Answer: B - Something You Have

Cyber Security

Threats & Web Vulnerabilities

1. How can developers effectively prevent SQL Injection (SQLi) vulnerabilities in application backends?

- A. By filtering HTML tags in input fields
- B. By using Parameterized Queries / Prepared Statements
- C. By disabling JavaScript on client browsers
- D. By encrypting the database password with MD5

Correct Answer: B - By using Parameterized Queries / Prepared Statements

2. What type of attack involves an attacker injecting malicious scripts into trusted websites that execute in victim browsers?

- A. Cross-Site Scripting (XSS)
- B. Cross-Site Request Forgery (CSRF)
- C. Man-in-the-Middle (MitM)
- D. Buffer Overflow

Correct Answer: A - Cross-Site Scripting (XSS)

3. What operational characteristic defines Ransomware malware?

- A. Silently monitoring user keystrokes for credentials
- B. Encrypting victim data and demanding payment for the decryption key
- C. Turning target devices into DDoS botnet zombies
- D. Overheating physical GPU components

Correct Answer: B - Encrypting victim data and demanding payment for the decryption key

Cyber Security

Network & Defensive Architecture

1. What is the foundational core philosophy of the Zero Trust Security Architecture?

- A. Trust any connection that originates from inside the internal corporate network
- B. Never trust, always verify every request regardless of origin
- C. Trust devices that pass initial perimeter firewall inspection
- D. Rely exclusively on strong perimeter network defenses

Correct Answer: B - Never trust, always verify every request regardless of origin

2. What feature distinguishes Next-Generation Firewalls (NGFW) from traditional packet-filtering firewalls?

- A. Deep Packet Inspection (DPI) and Application Layer (Layer 7) awareness
- B. Ability to block physical cable connections
- C. Support for basic port and IP filtering only
- D. Elimination of NAT configuration requirements

Correct Answer: A - Deep Packet Inspection (DPI) and Application Layer (Layer 7) awareness

3. Which security tool aggregates, correlates, and analyzes log data across an enterprise to detect real-time threats?

- A. SIEM (Security Information and Event Management)
- B. DHCP Server
- C. Reverse Proxy
- D. SAN Array

Correct Answer: A - SIEM (Security Information and Event Management)

4. What software agent is deployed directly on user workstations to monitor, detect, and isolate malicious host activity?

- A. EDR (Endpoint Detection and Response)
- B. NTP Client
- C. DNS Resolver
- D. Load Balancer

Correct Answer: A - EDR (Endpoint Detection and Response)

NCW

NCW Principles & Doctrine

1. What is the primary objective of Network-Centric Warfare (NCW) doctrine?

- A. Massing physical troops on the frontline
- B. Translating an information advantage into operational superiority by robustly networking sensors, commanders, and effectors
- C. Replacing all manned military units with fully autonomous platforms
- D. Centralizing all tactical decisions strictly within regional headquarters

Correct Answer: B - Translating an information advantage into operational superiority by robustly networking sensors, commanders, and effectors

2. According to NCW tenets, what key operational benefit does shared situational awareness enable among distributed forces?

- A. Manual paper mapping
- B. Self-synchronization and accelerated decision cycles (OODA loop)
- C. Elimination of wireless communications requirement
- D. Single-point-of-failure operations

Correct Answer: B - Self-synchronization and accelerated decision cycles (OODA loop)

3. What does the military acronym C4ISR stand for in network-centric operational domains?

- A. Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance
- B. Cyber, Central, Coding, Cryptography, Integrated, Systems, and Radar
- C. Combat, Counter, Control, Defense, Information, Security, and Reconnaissance
- D. Cooperative, Centralized, Command, Communications, Integrated, Radar, and Sensors

Correct Answer: A - Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance

NCW

Tactical Data Links & Battlespace Grids

1. Which secure, jam-resistant tactical data link protocol is widely used for real-time situational data exchange among air, sea, and land assets?

- A. Link 16
- B. HTTP/2
- C. FTP
- D. X.25

Correct Answer: A - Link 16

2. In Net-Centric Operations, what is a Common Operational Picture (COP)?

- A. A static satellite photograph published daily
- B. A single identical display of relevant operational information shared across echelons
- C. A encrypted video stream from a single ground unit
- D. An offline tactical map stored in local storage

Correct Answer: B - A single identical display of relevant operational information shared across echelons

3. What component forms the 'Sensor-to-Shooter' grid in NCW architecture?

- A. Isolated radar systems connected directly to local artillery without network links
- B. Networked integration connecting reconnaissance sensors directly to command nodes and weapon systems for fast target engagement
- C. Manual voice radio relay over analog HF lines
- D. Unencrypted civilian cellular towers

Correct Answer: B - Networked integration connecting reconnaissance sensors directly to command nodes and weapon systems for fast target engagement

NCW

Net-Centric Technologies & Resilience

1. What architecture standard allows Software Defined Radios (SDR) in net-centric systems to dynamically load waveforms on reconfigurable hardware?

- A. Software Communications Architecture (SCA)
- B. OpenFlow Protocol
- C. REST API
- D. POSIX Standard

Correct Answer: A - Software Communications Architecture (SCA)

2. What primary vulnerability arises from a military force becoming heavily reliant on NCW operations?

- A. Decreased bandwidth efficiency
- B. High susceptibility to electronic warfare, cyber attacks, and satellite communications jamming
- C. Inability to deploy modern computer systems
- D. Excessive reliance on physical mail delivery

Correct Answer: B - High susceptibility to electronic warfare, cyber attacks, and satellite communications jamming

3. What DoD framework provides the enterprise network backplane supporting net-centric military operations?

- A. Global Information Grid (GIG) / DODIN
- B. Public Internet Backbone
- C. ARPANET
- D. Local Loop Unbundling

Correct Answer: A - Global Information Grid (GIG) / DODIN

4. What role does Satellite Communications (SATCOM) play in geographically dispersed NCW force deployments?

- A. Providing short-range line-of-sight voice communications only
- B. Providing high-bandwidth beyond-line-of-sight (BLOS) data connectivity connecting field units to strategic C2 hubs
- C. Replacing local radar processing servers
- D. Providing physical shielding against electromagnetic pulses

Correct Answer: B - Providing high-bandwidth beyond-line-of-sight (BLOS) data connectivity connecting field units to strategic C2 hubs

Networking

OSI & TCP/IP Protocols

1. At which OSI layer does logical IP addressing and path determination occur?

- A. Data Link Layer (Layer 2)
- B. Network Layer (Layer 3)
- C. Transport Layer (Layer 4)
- D. Session Layer (Layer 5)

Correct Answer: B - Network Layer (Layer 3)

2. In TCP/IP networking, what is the exact 4-step message sequence exchanged during standard DHCP address lease allocation?

- A. Discover, Offer, Request, Acknowledge (DORA)
- B. Query, Response, Accept, Confirm
- C. Syn, Syn-Ack, Ack, Fin
- D. Hello, Request, Lease, Confirm

Correct Answer: A - Discover, Offer, Request, Acknowledge (DORA)

3. Which Transport Layer protocol provides reliable, connection-oriented data transmission using sequence numbers and acknowledgments?

- A. UDP
- B. TCP
- C. ICMP
- D. IPsec

Correct Answer: B - TCP

Networking

Routing & Switching

1. Which interior gateway protocol (IGP) uses Dijkstra's Shortest Path First (SPF) algorithm to build link-state topology tables?

- A. RIP
- B. OSPF
- C. BGP
- D. EIGRP

Correct Answer: B - OSPF

2. What standard protocol IEEE tag format is used to identify VLAN membership inside Ethernet frame headers?

- A. IEEE 802.11ax
- B. IEEE 802.1Q
- C. IEEE 802.3af
- D. IEEE 802.1X

Correct Answer: B - IEEE 802.1Q

3. Which Exterior Gateway Protocol (EGP) is responsible for exchanging routing information between different Autonomous Systems across the global Internet?

- A. OSPF
- B. BGP (Border Gateway Protocol)
- C. IS-IS
- D. RIPv2

Correct Answer: B - BGP (Border Gateway Protocol)

Networking

Modern Network Technologies & Wireless

1. In Software-Defined Networking (SDN), what fundamental structural change is made to traditional router/switch design?

- A. Merging the Data Plane and Management Plane into physical chips
- B. Decoupling the Control Plane from the underlying Data (Forwarding) Plane
- C. Removing hardware network interface cards
- D. Replacing optical fiber with coaxial cables

Correct Answer: B - Decoupling the Control Plane from the underlying Data (Forwarding) Plane

2. What multi-user transmission technology introduced in Wi-Fi 6 (802.11ax) enables efficient subcarrier channel sharing among simultaneous clients?

- A. OFDMA (Orthogonal Frequency-Division Multiple Access)
- B. CSMA/CD
- C. WEP Encryption
- D. FHSS

Correct Answer: A - OFDMA (Orthogonal Frequency-Division Multiple Access)

3. Which version of SNMP added strong user authentication and payload encryption capabilities?

- A. SNMP v1
- B. SNMP v2c
- C. SNMP v3
- D. SNMP v4

Correct Answer: C - SNMP v3

4. When a DNS resolver receives a referral to another authoritative name server rather than the final answer, what query type is being executed?

- A. Recursive Query
- B. Iterative Query
- C. Reverse Lookup
- D. Zone Transfer

Correct Answer: B - Iterative Query